

Shower project

Team OH-J

Element H: Prototype Testing and Data Analysis

In the boys dorms the dorm residents need a place to put their toiletries because there is not a built-in shelf in the communal showers and setting things on the floor is unsanitary. Solving the problem can help the dorm students place their toiletries somewhere without having to set it down on the ground. The problem is that there is nowhere to hang your shampoo and that leads to people putting it on the ground.

We can test for stability by doing different tests to see if the caddy can hang on the wall or be able to withstand being bumped into. The caddy will be close to the wall but will be suspended by metal hooks to go over the wall. The caddy will be tested to see if the measurements are right of length, width and height. We will do a durability test if the box is strong enough to hit the ground and not break. Capacity test on how many soap bottles can fit inside. 2 medium sized bottles and 2 small bottles are able to fit inside. The estimated cost of our prototype is \$15 and that is because it does not take that long to make but the material is durable and is water resistant. We will do a portability test and the caddy will be held with one hand and can be moved around easily without restrictions.

Stability test

1. Install the caddy.
2. Record observations on how level the caddy hangs.
3. Add 5 toiletry bottles to the caddy.
4. Record observations on how level the caddy hangs.

5. Repeat steps 1-4, 2 additional times.

Elevated off floor

1. Install the caddy.
2. Measure the distance from the floor.
3. Record data.
4. Repeat steps 1-3, 2 additional times.

Size Test

1. Measure the length, width, and height.
2. Record data.

Durability

1. Install caddy
2. Test to see if caddy can withstand being dropped
3. Record data
4. Repeat steps 2, 2 additional times

Capacity

1. Install caddy
2. Test to see if 5 bottles can be placed inside
3. Record data
4. Repeat steps 2-3, 2 additional times.

Portability

1. Install caddy
2. Test to see if box can be carried with one hand
3. Record data
4. Repeat steps 2-3, 2 additional times.

Ms. Z was the one that went over our testing plan and allowed us to move on to the future steps.